Engine Model(s): PT6A

76-10-00

PROPELLER REVERSING LINKAGE - MAINTENANCE PRACTICES

1. General

A. Maintenance personnel should make reference to the INTRODUCTION section and Chapter 70-00-00 STANDARD PRACTICES of this manual to familiarize themselves with general procedures.

B. Install suitable protective caps/covers over all disconnected tubes/lines and component openings.

C. Lockwire shall comply with specification AMS5687, heat and corrosion resistant steel wire MS9226-03, which is 0.025 inch diameter, and will not be specified in instructions.

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name
PWC04-001	Grease, Synthetic
PWC05-061	Cloth, Abrasive Coated
PWC11-027	Solvent, Petroleum
PWC11-031	Cleaner, Engine

3. Special Tools

The special tools listed below are used in the following procedures.

Tool No.	Name
PWC37548	Drift

4. Fixtures, Equipment and Supplier Tools

Not Applicable

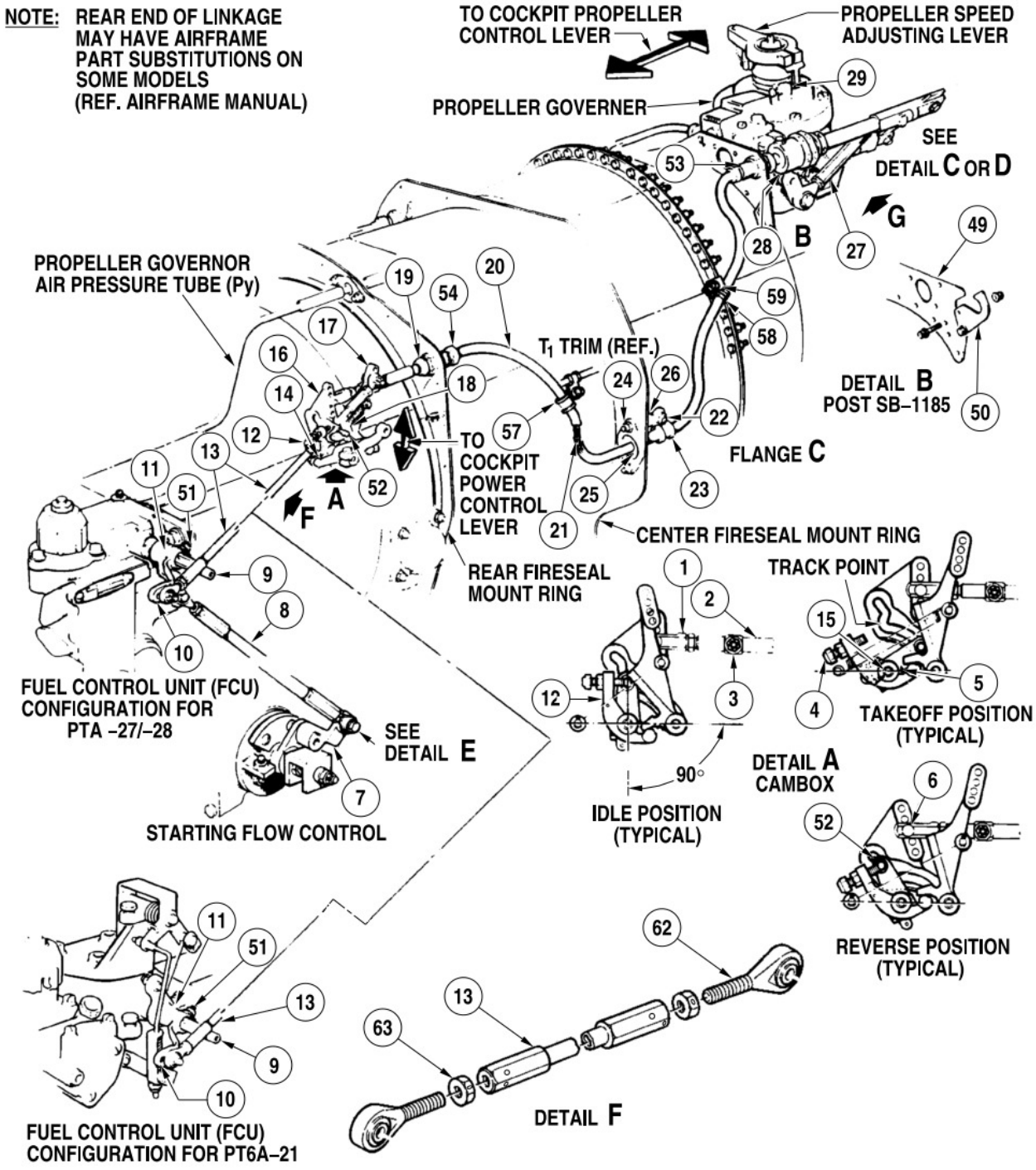
5. Removal/Installation

A. Removal (Ref. Fig. 201)

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Figure 201 Removal/Installation of Propeller Reversing Linkage

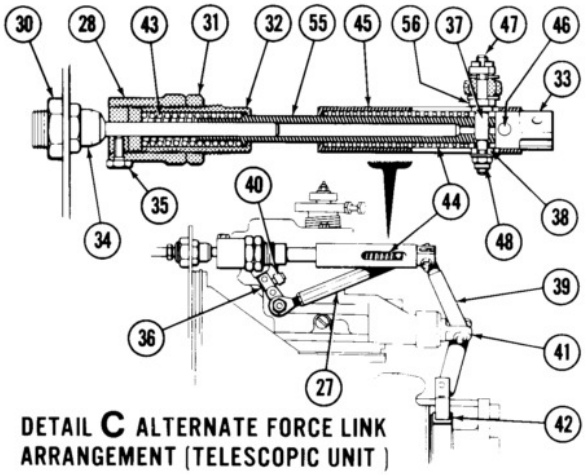
(SHEET 1 OF 5)



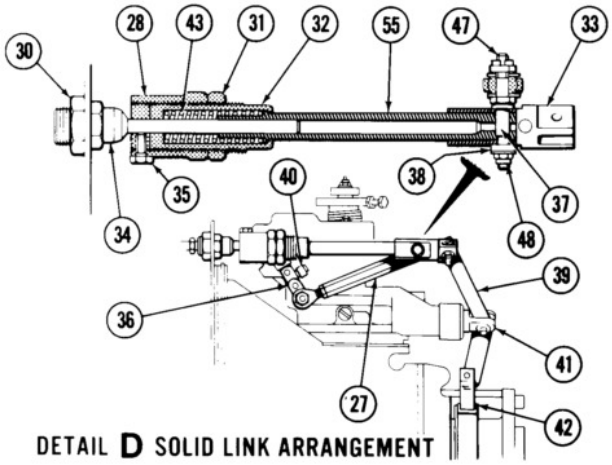
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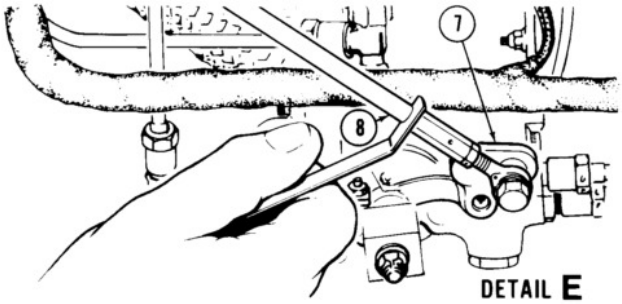
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DETAIL C ALTERNATE FORCE LINK ARRANGEMENT (TELESCOPIC UNIT)



DETAIL D SOLID LINK ARRANGEMENT

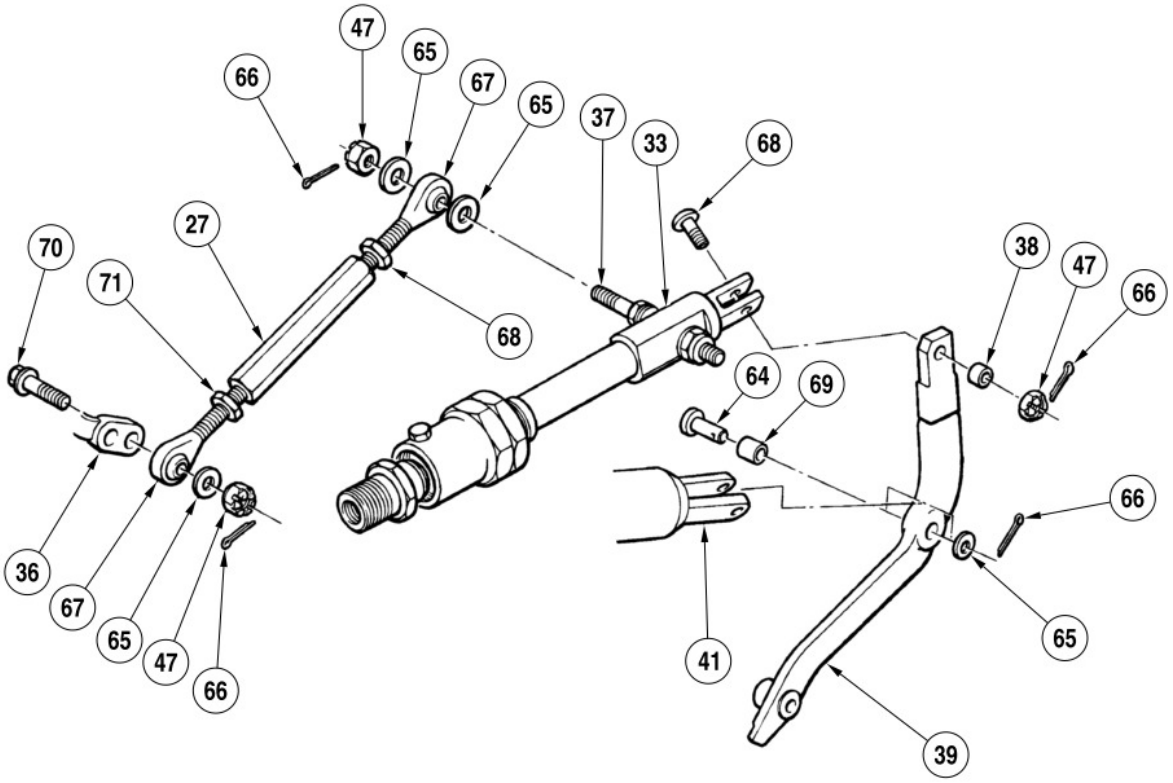


DETAIL E

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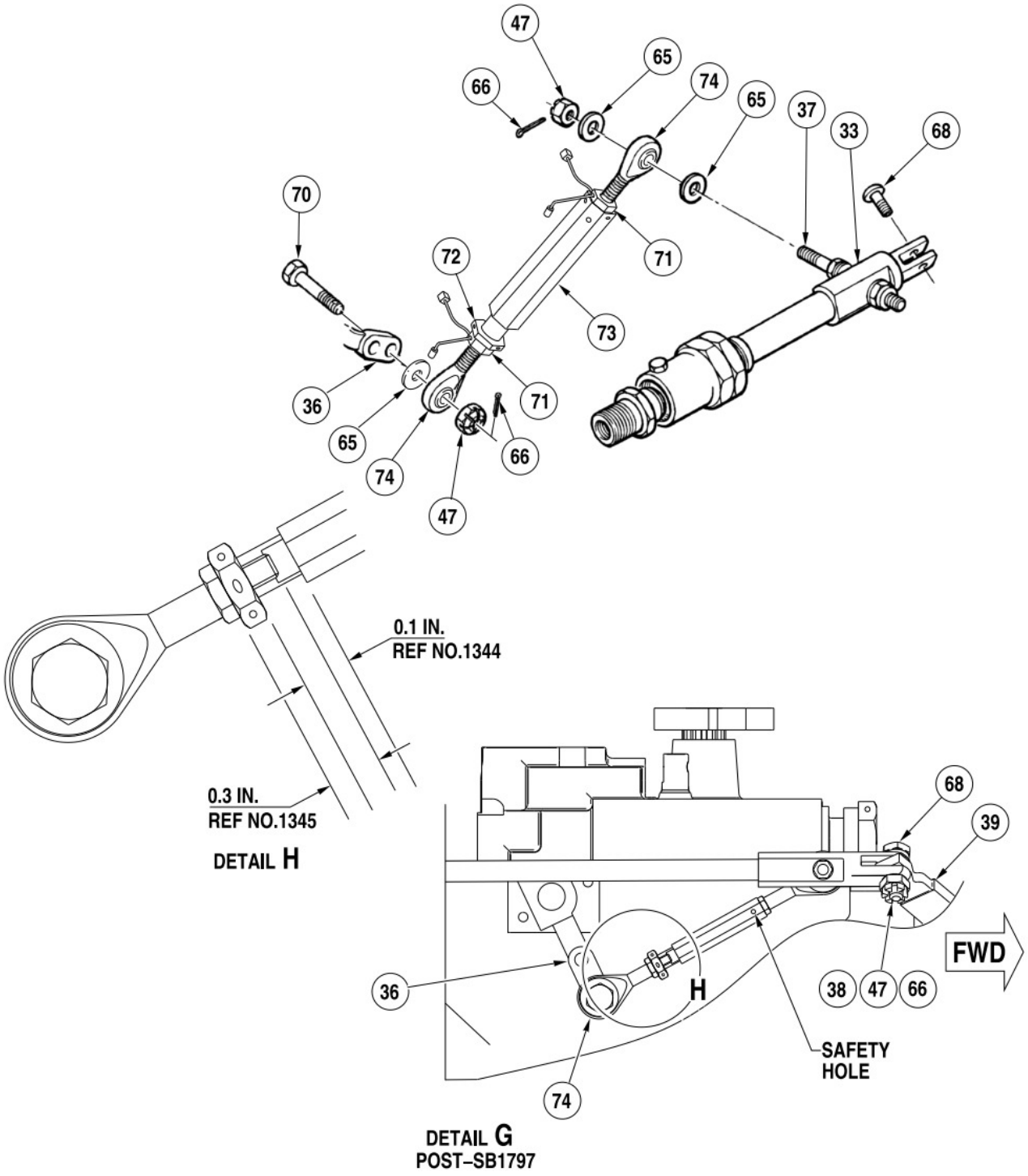


DETAIL G
PRE-SB1797

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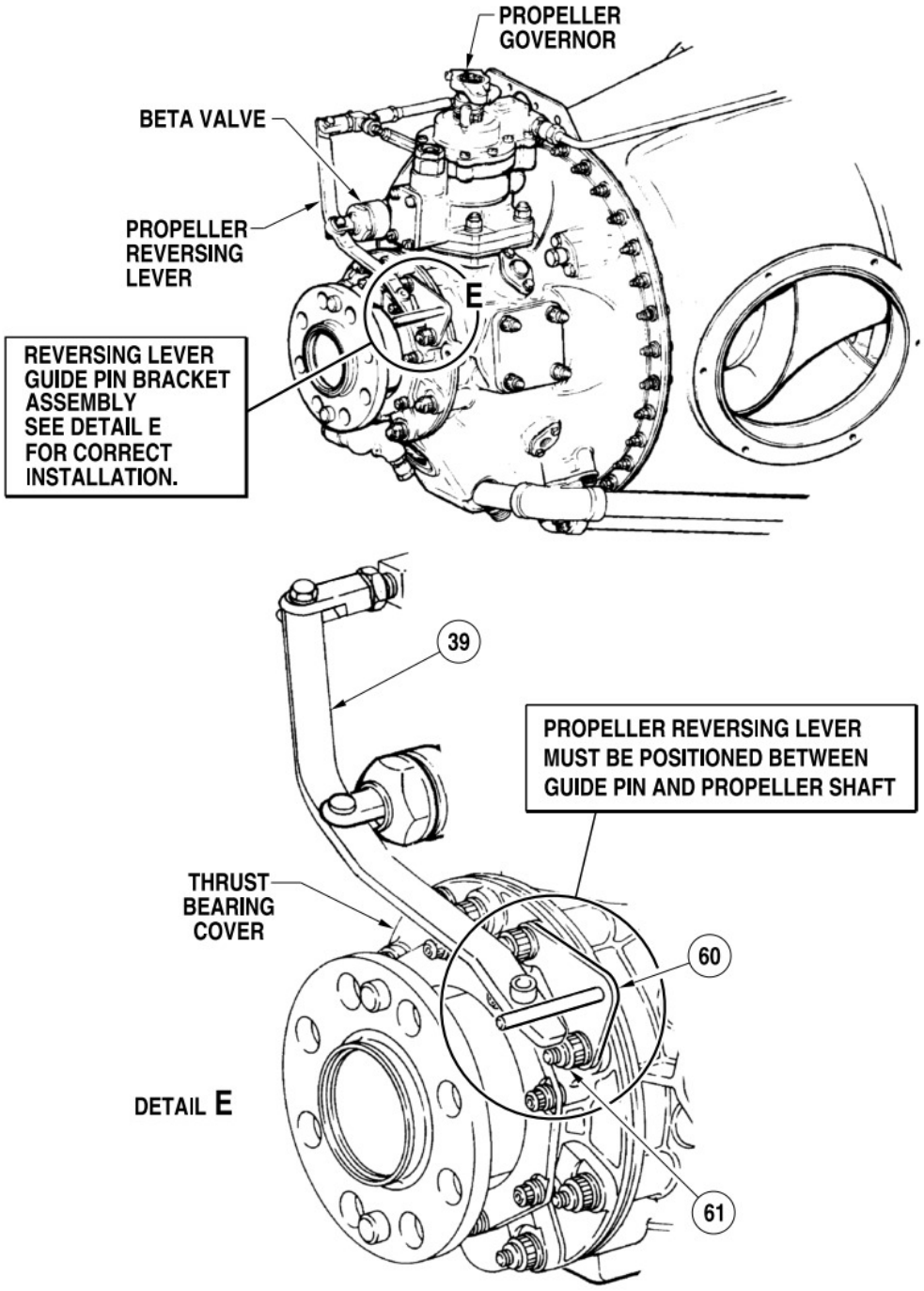
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1. Rod End Clevis (Rear)
2. Wire Rope Terminal
3. Clamping Bolt (Rear)
4. Bolt - Idle Deadband
5. Reversing Control Follower Lever
6. Clevis Pin
7. Starting Control Lever (PT6A-27/28 only)
8. Starting Control Rods (PT6A-27/28 only)(Pre-SB1152)
9. FCU Arm Extension
10. FCU Arm
11. FCU Arm Serrated Spacer
12. Actuating Lever
13. FCU Control Rod
14. Control Lever Mounting Bracket
15. Grooved Pin
16. Propeller Control Cam
17. FCU Actuating Lever
18. Extension Spring
19. Rear Swivel Joint
20. Wire Rope Casing
21. Wire Rope
22. Self-Locking Nut
23. Loop Clamp
24. Self-Locking Nut and Bolt
25. Retaining Plate and Seals
26. Bracket
27. Propeller Governor Interconnect Rod (**Pre-SB1797**)
28. Low Pitch Stop Adjuster
29. Propeller Speed Adjusting Lever Maximum Stop
30. Locknut
31. Locknut
32. Low Pitch Adjuster Stop
33. Rod End Clevis (Front)
34. Front Swivel Joint
35. Lockbolt
36. Propeller Governor Reset Arm
37. Clamping Bolt (Front)
38. Spacer
39. Propeller Reversing Lever
40. Airbleed Link Maximum Stop
41. Beta Control Valve Clevis
42. Carbon Block
43. Compression Spring (Pre-SB1129)
44. Compression Spring
45. Terminal Housing
46. Clevis Retaining Bolt

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47. Castellated Nut
48. Self-Locking Nut
49. Front Lifting Bracket
50. Retaining Plate
51. Adjusting Screw
52. Cam Follower Pin
53. Coupling Nut
54. Coupling Nut
55. Wire Rope Terminal
56. Terminal Sleeve
57. Loop Clamp
58. Loop Clamp
59. Angle Bracket
60. Guide Pin Bracket Assembly
61. Self-Locking Nut
62. Rod End Connector
63. Locknut
64. Pin
65. Washer
66. Cotter Pin
67. Rod End Connector (**Pre-SB1797**)
68. Bolt
69. Sleeve Bushing
70. Bolt
71. Nut
72. Reset Arm Adjuster (**Post-SB1797**)
73. Internally Threaded Reset Arm Adjuster (**Post-SB1797**)
74. Rod End Connector (**Post-SB1797**)

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NOTE: For procedure associated with disconnecting propeller reversing linkage when removing the engine power section, refer to 72-00-00.

(1) Disconnect airframe cockpit lever linkage from grooved pin (15) of cam box at flange G and propeller governor (Ref. Airframe Manufacturer's Manual.)

(2) Remove rear linkage:

(a) For Pre-SB1658/Pre-SB1666 Engines: Remove cotterpins, castellated nuts, washers and bolts that attach FCU control rod (13) to FCU arm (10) and FCU actuating lever (17).

(b) For Post-SB1658/Post-SB1666 Engines: Remove cotterpins, castellated nuts and bolts that attach FCU control rod (13) to FCU arm (10) and FCU actuating lever (17).

(c) Remove FCU control rod.

(d) Remove cotterpin, washer and clevis pin (6) that secure rear rod end clevis (1) to propeller control cam (16).

(e) Loosen self-locking nut on rear clamping bolt (3). Disengage terminal and rod end assembly (2) from wire rope (21) by turning counterclockwise to free rope from clamping bolt internal thread.

(f) Remove cam box by removing two self-locking nuts at flange G, and bolt that secure assembly to angle bracket on accessory gearbox housing. Remove assembly.

(g) Disassemble cam box:

1 Remove cotterpins and washers that secure extension spring (18) to FCU actuating lever (17) and bracket.

2 Remove cotterpin, washer and flat headed pin that secure propeller control cam (16) and FCU actuating lever (17) to bracket.

3 Remove cotterpin, washer and cam follower pin (52) that secure propeller reversing control follower lever (5) to propeller control cam (16).

4 Loosen bolt that secure reversing control follower lever to grooved pin (15). Withdraw grooved pin from bracket and remove follower lever and FCU actuating lever (12).

(h) Remove front linkage:

1 Loosen and unscrew coupling nut (53) at front end of wire rope casing (20) from front swivel joint (34).

2 Remove cotterpin, castellated nut, spacer and bolt securing front rod end clevis (33) to propeller reversing lever (39).

3 Remove cotterpin, washer and straight pin securing propeller reversing lever (39) to Beta control valve clevis (41). Remove reversing lever.

4 **Pre-SB1797**:

a Remove the cotter pin (66), castle nut (47), washer (65) and the bolt (70) that attach rear end of rod assembly (27) to the airbleed link reset lever (36) on the propeller governor.

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b Remove the cotter pin (66), castle nut (47) and the two washers (65) that attach the upper end of the interconnect rod (27) to the front clamping bolt (37). Remove the rod assembly (27) and the connectors (67).

5 **Post-SB1797**:

a Remove the cotter pin (66), the castle nut (47), the washer (65) and the bolt (70) that attach the rear end of reset arm adjuster assembly to the airbleed link reset lever (36).

b Remove the cotter pin (66), the castle nut (47) and the two washers (65) that attach the forward end of the reset arm adjuster assembly to the front clamping bolt (37). Remove the reset arm adjuster assembly and the connectors (74).

6 For Pre-SB1185 Engines: Remove wire rope and front end of linkage:

a Remove locknut (30) securing front swivel joint (34) to front lifting bracket (49).

b Remove swivel joint (see Detail D or C as applicable) complete with wire rope terminal, low pitch stop adjuster (28), low pitch adjuster stop (32), compression spring (43) (if fitted) and wire rope (21).

NOTE: Spring (43) is omitted on some installations (Ref. IPC for usage.)

7 For Post-SB1185 Engines: Remove wire rope and front end of linkage:

a At front lifting bracket (49, Detail B) remove self-locking nut and bolt nearest engine and loosen remaining nut securing retaining plate (50) to lifting bracket.

b Loosen locknut (30) securing front swivel joint to lifting bracket.

c Pivot retaining plate outwards and remove swivel joint and attached control linkage and wire rope.

8 Remove self-locking nut (22), washer and bolt securing wire rope loop clamp (23) to bracket (26) at center fireseal.

9 Remove self-locking nut, bolt and loop clamp from similar loop clamp on T1 trim harness probe.

10 Remove self-locking nut, bolt and the loop clamp (58) from the bracket (59) at flange C. Disconnect the coupling nut (54) at rear end of the wire rope casing, from the rear swivel joint (19).

11 Remove two self-locking nuts (24) and bolts securing seals, seal retaining plates and seal retaining plate insulation (25) to center fireseal mount ring. Remove seal and retaining plates from rear end of wire rope casing.

12 Withdraw wire rope casing forward through center fireseal mount ring.

13 Remove locknut securing rear swivel joint to rear fireseal mount ring and support bracket. Remove swivel joint.

14 Disassemble push/pull control link arrangement (Ref. Detail D):

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a Remove self-locking nut (48) and spacer (38) from front clamping bolt (37).

b Remove wire rope (21) from front linkage by rotating rope counterclockwise to disengage rope from internal thread on clamping bolt (37). Place wire rope in a plastic bag to prevent pick-up of dirt and other foreign matter.

c Remove clamping bolt (37) from wire rope terminal and rod end clevis (33).

d Remove locknut (31) securing low pitch adjuster stop (32) to low pitch stop adjuster (28).

e Unscrew adjuster stop from stop adjuster, and remove compression spring (43, if fitted, Pre-SB1129) and wire rope terminal (55).

f Remove lockbolt (35) securing low pitch stop adjuster to front swivel joint.

15 On engines incorporating force link arrangement (Telescopic Unit) disassemble (Ref. Detail C):

a Remove self-locking nut (48) and spacer (38) from front clamping bolt (37).

b Remove wire rope (21) from front linkage by rotating rope counterclockwise to disengage rope from internal thread of clamping bolt. Place wire rope in plastic bag to prevent pick-up of dirt and other foreign matter.

c Remove clamping bolt (37) and slide terminal housing (45) off wire rope terminal (55).

d Remove cotterpin, castellated nut and bolt (46) securing rod end clevis (33) to terminal housing. Remove rod end clevis, terminal sleeve (56) and compression spring (44) from housing.

e Remove locknut (31) securing low pitch adjuster stop (32) to low pitch stop adjuster (28).

f Unscrew adjuster stop from stop adjuster and remove compression spring (43, Pre-SB1129) and wire rope terminal (55).

g Remove lock bolt (35) securing low pitch stop adjuster to front swivel joint.

(3) Remove guide pin bracket assembly:

(a) Remove two nuts (61).

(b) Remove guide pin bracket assembly (60).

B. Installation (Ref. Fig. 201)

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CAUTION: THE FOLLOWING ASSEMBLY INSTRUCTIONS ARE GIVEN PRIMARILY TO SATISFY SHIPPING REQUIREMENTS AND THE RESULTANT RIGGING MAY DIFFER QUITE CONSIDERABLY FROM WHAT IT SHOULD BE WHEN THE ENGINE IS INSTALLED IN THE AIRFRAME. AT ENGINE INSTALLATION BY THE AIRFRAME MANUFACTURER THE LINKAGE IS ADJUSTED TO SUIT A PARTICULAR AIRFRAME REQUIREMENT. THIS ADJUSTMENT MAY INCLUDE ALTERING ROD LENGTHS, MOVING CLEVIS PINS TO DIFFERENT HOLES AND, IF NECESSARY, SELECTING A DIFFERENT CAM TO PROVIDE THE REQUIRED POWER RESPONSE CHARACTERISTICS. THE FINAL RIGGING ARRANGEMENT IS PART OF THE AIRCRAFT CERTIFICATION REQUIREMENT AND IS DETAILED IN THE APPROPRIATE AIRCRAFT MAINTENANCE MANUAL. THEREFORE, WHEN AN ENGINE IS REINSTALLED IN AN AIRCRAFT AFTER OVERHAUL OR REPAIR, THE ENGINE SHOULD BE RIGGED IN ACCORDANCE WITH THE RELEVANT AIRCRAFT MAINTENANCE MANUAL AND NOT IN ACCORDANCE WITH THE APPLICABLE P&WC MAINTENANCE OR OVERHAUL MANUAL.

(1) Install guide pin bracket assembly:

(a) Install guide pin bracket assembly (60).

(b) Install nuts (61) and torque 145 to 165 lb.in.

(2) Install and adjust front linkage:

(a) Install front swivel joint on front lifting bracket:

1 For Pre-SB1185 Engines: Pass threaded end of swivel joint (34) through hole in lifting bracket at flange A and secure with locknut (30). Tighten nut, torque to 95 to 105 lb.in., and lockwire.

2 For Post-SB1185 Engines: Install threaded end of swivel joint (34) in slotted hole of front lifting bracket (49). Pivot retaining plate (50) over slot and secure with bolt and self-locking nut. Tighten retaining plate nuts and torque to 36 to 40 lb.in. Install locknut (30) on swivel joint. Tighten nut, torque to 95 to 105 lb.in., and lockwire.

(b) Install low pitch stop adjuster (28) on swivel joint and secure with lockbolt (35). Tighten bolt, torque to 32 to 36 lb.in., and lockwire.

(c) Install compression spring (43) (if fitted) on front section of swivel joint.

NOTE: Spring (43) is omitted on some installations (Ref. IPC. for usage).

(d) Install low pitch adjuster stop (32) onto wire rope terminal (55). Slide wire rope terminal over swivel joint and screw low pitch adjuster stop into stop adjuster (28) to obtain between 1.00 and 1.25 inch travel on wire rope terminal. Screw locknut (31) onto adjuster stop and tighten fingertight only.

NOTE: Locknut must be torqued and lockwired after final adjustments (Ref. Step (e), following).

(e) Assemble push/pull control linkage arrangement (Ref. Detail D):

1 Assemble front rod end clevis (33) on wire rope terminal (55) and insert front clamping

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bolt (37) through holes in terminal and clevis. Ensure wire rope threaded hole in bolt is lined up correctly and screw wire rope into bolt thread. Check that wire rope is fully engaged by inserting wire in inspection hole in clevis (33).

NOTE: Installation of wire rope into terminal at this stage only applies when rear linkage and wire rope casing is already installed. If rear linkage and wire rope casing is not installed do not assemble wire rope.

2 Secure clamping bolt (37) with spacer (38) and self-locking nut (48). Tighten nut and torque to 10 to 15 lb.in.

(f) On engine incorporated force link arrangement (Telescopic Unit) (Ref. Detail C):

1 Slide terminal housing (45) over wire rope terminal (55).

2 Install compression spring (44) into housing, followed by terminal sleeve (56) and rod end clevis (33).

3 Secure rod end clevis to terminal housing with bolt (46) and castellated nut. Tighten nut, torque 24 to 36 lb.in., and lock with cotterpin.

4 Install wire rope clamping bolt (37) through slotted hole in side of terminal housing and through hole in terminal sleeve (56).

5 Screw wire rope into clamping bolt internal thread and ensure end of rope is visible through inspection hole in housing. Install spacer (38) and self-locking nut (48). Tighten nut and torque 10 to 15 lb.in.

NOTE: Installation of wire rope into terminal at this stage only applies when rear linkage and wire rope casing is already installed. If rear linkage and wire rope casing is not installed do not assemble wire rope.

(g) With the propeller feathered and the carbon block (42) put against the rear face of the propeller feedback ring, adjust the low pitch adjuster stop (32) with the linkage pulled fully forward, the beta valve clevis slot end is flush with the valve capnut. Tighten the locknut (31) 150 to 250 lbf.in. Attach the locknut to the stop adjuster (28) with the lockwire or safety wire.

(h) **Pre-SB1797**:

CAUTION: THE LOWER END OF THE PROPELLER REVERSING LEVER (39) IS MACHINED WITH A STEPPED NOTCH. MAKE SURE THE STEPPED NOTCH AT THE END OF THE PROPELLER REVERSING LEVER (39) IS UNDER THE PIN IN THE REVERSING LEVER GUIDE PIN BRACKET ASSEMBLY (60) (REF. FIG. 201, DETAIL E).

1 Install the sleeve bushing (69) with the drift (PWC37548) into the hole at the center of the lever (39). Position the lever into the fork of the beta valve plunger (41). Attach the lever with the pin (64), washer (65) and the cotterpin (66).

2 Install the washer (65), the rod assembly (27), the rod end connector (67), the washer (65) and the castle nut (47) on the wire rope clamp (37). Torque the castle nut 24 to 36 lbf.in. Lock the nut with the cotter pin (66).

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3 Install the sleeve spacer (38) in the upper hole of the propeller lever (39). Engage the upper end of the lever (39) in the push-pull wire rope front terminal clevis (33). Install the bolt (68) through the clevis and the lever. Attach the clevis and the lever with the castle nut (47). Torque the castle nut 24 to 36 lbf.in. Lock the castle nut with cotter pin (66).

4 Position the interconnect linkage fully forward against the low pitch stop and position airbleed link (36) fully forward against the max. stop. Align the rear connector (67) of the rod assembly (27) with the airbleed link (36). Make sure that the center of the connector hole is approximately 1/32 inch to the rear of center of the outer hole in the link (36). If necessary, adjust the length of the rod assembly (27).

5 To make sure that the two connectors (67) of the rod assembly (27) are in safety, put an end of a stiff piece of lockwire or safety wire in safety holes in the rod. If the lockwire or safety wire does not pass through the holes, then the connectors are in safety. Torque the two nuts (47) 12 to 18 lbf.in. Install the lockwire or safety wire on the connectors (67) and the nut (47) to the rod (27).

NOTE: Final adjustment of the low pitch stop position and rod assembly can only be achieved with a propeller installed.

6 Attach the rear connector (67) to the airbleed link (36) with the bolt (70), washer (65) and the castle nut (47). Torque the nut 24 to 36 lbf.in. Lock the nut with the cotter pin (66).

(i) **Post-SB1797**:

CAUTION: THE LOWER END OF THE PROPELLER REVERSING LEVER (39) IS MACHINED WITH A STEPPED NOTCH. MAKE SURE THE STEPPED NOTCH AT THE END OF THE PROPELLER REVERSING LEVER (39) IS UNDER THE PIN IN THE REVERSING LEVER GUIDE PIN BRACKET ASSEMBLY (60) (REF. FIG. 201, DETAIL E).

1 Install the sleeve bushing (69) with the drift (PWC37548) into the hole at the center of the lever (39). Position the lever into the fork of the beta valve plunger (41). Attach the lever with the pin (64), washer (65) and the cotterpin (66).

2 Install the washer (65), the reset arm adjuster (72), the internally threaded reset arm adjuster (73), the connector (74), the washer (65) and the castle nut (47) on the wire rope clamp (37). Torque the castle nut 24 to 36 lbf.in. Lock the castle nut with the cotter pin (66).

3 Install the sleeve spacer (38) in the upper hole of the propeller lever (39). Engage the upper end of the lever (39) in the push-pull wire rope front terminal clevis (33). Install the bolt (68) through the clevis and the lever. Attach the clevis and the lever with the castle nut (47). Torque the castle nut 24 to 36 lbf.in. Lock the castle nut with cotter pin (66).

4 Position the interconnect linkage fully forward against the low pitch stop. Position the airbleed link (36) fully forward against the max. stop. Align the rear connector (74) with

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the airbleed link (36).

5 Turn the rod end connector (74) into the reset arm adjuster (72) to get 0.3 in. dimension from rear bolt head face shown on Detail H (REF. NO. 1345, Fits and Clearances).

NOTE: Three complete turns of the reset arm adjuster (72) are equivalent to 0.1 in.

6 Attach the rear connector (74) to the link (36) with the bolt (70), washer (65) and the castle nut (47). Torque the castle nut 24 to 36 lbf.in. Lock the castle nut with the cotter pin (66).

7 Set the airbleed link (36) against the stop while in free position. Adjust the gap between the reset arm adjuster (72) and the internally threaded reset arm adjuster (73) 0.1 in. max. (Ref. Detail H) (REF. NO. 1344, Fits and Clearances).

8 Do a visual inspection of the internally threaded reset arm adjuster (73) inspection hole to make sure that the threads of rod end connector (74) extend fully into the adjuster.

9 To make sure that the two connectors (74) are in safety, put an end of a stiff piece of lockwire or safety wire in safety holes in rod. If the lockwire or safety wire does not pass through the holes, then the connectors are in safety.

10 Torque the two nuts (71) 12 to 18 lbf.in. Install the lockwire or safety wire on the connectors (74) and the nut (71) to the reset arm adjuster (72) and the internally threaded reset arm adjuster (73).

NOTE: Final adjustment of the low pitch stop position and adjusting rod assembly can only be achieved with a propeller installed.

(3) Install and adjust rear linkage:

NOTE: 1. Rigging of the rear linkage will vary considerably dependent on airframe installations and type of control cam (Quiet Taxi, Linear Beta, etc.) being used. Airframe part substitutions are made on some installations (Ref. Airframe Manufacturer's Manual).

NOTE: 2. Post-SB1486 and Post-SB1492 engines incorporate an FCU actuating lever featuring a positive return ramp that will drive towards idle in the event of rod end seizure of the control linkage.

(a) Assemble FCU actuating lever (17) and propeller control cam (16) to control lever mounting bracket (14) and insert flat headed pin through front holes in bracket. Secure pin with washer and cotterpin.

(b) Assemble actuating lever (12) and reversing control cam follower lever (5) to control lever mounting bracket (14). Install grooved pin (15) through rear holes of bracket and two levers. Tighten bolt on follower lever (5), torque 32 to 36 lb.in., and lockwire.

(c) Install fork end of follower lever over propeller control cam (16) and insert cam follower pin (52) through holes in fork end of lever and through cam slot. Secure with washer and cotterpin.

(d) Temporarily secure angle bracket to control lever bracket (14) with single bolt. Do not torque bolt at this stage.

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(e) Locate control lever mounting bracket assembly and bracket at flange G of accessory gearbox and secure to gearbox studs with two self-locking nuts and to gearbox boss with two bolts. Tighten nuts and bolts, torque to 36 to 40 lb.in., and lockwire bolts.

(f) Install extension spring (18) between post on control lever mounting bracket (14) and post of FCU actuating lever (17) and secure with plain washers and cotterpins.

(g) Install and adjust rear linkage:

1 Rotate FCU arm (10) fully counterclockwise and then slowly clockwise until the FCU governor cam pick-up point is felt.

NOTE: Initial movement of the FCU arm from the fully counterclockwise position is free from resistance. The pick-up point is recognized when a slight resistive force is encountered.

2 Slacken FCU arm extension (9) and adjust serrated spacer (11) until FCU arm (10) is approximately 16 degrees below horizontal as the pick-up point is reached. The serrations on the FCU arm together with unequal number of serrations on both sides of spacer (11) provide adjustment in increments of 0.6 degrees. Tighten FCU arm extension (9), torque to 25 to 35 lb.in., and lockwire.

NOTE: Angle of 16 degrees is typical nominal setting. Refer to rigging instructions for particular airframe type for actual setting angle requirements.

3 Install and adjust idle deadband bolt (4) in actuating lever (12) to suit specific airframe installation and propeller control cam (Ref. Airframe Manufacturer's Manual for rigging requirements.)

4 Rotate power input lever (if fitted) or grooved pin (15) until cam follower pin (52) (headless shoulder pin) rests in track point of cam (16) (IDLE POSITION, Ref. Detail A).

5 With FCU arm (10) set approximately two degrees counterclockwise from pick-up point and rear linkage in idle position (step (3)(d), preceding), adjust length of FCU control rod (13) until holes at rod end connectors align with top hole of FCU actuating lever (17) and inner hole of FCU arm (10).

6 For **Pre-SB1658**/ **Pre-SB1666** Engines: Apply light grease (PWC04-001) to the rod end. Tighten rod end connector locknuts and install control rod. Temporarily attach with bolts, washers and castellated nuts.

7 For **Post-SB1658**/ **Post-SB1666** Engines: Apply light grease (PWC04-001) to the rod end. Tighten rod end connector locknuts and install control rod. Temporarily attach with bolts and castellated nuts.

8 Rotate power input lever (if fitted) or grooved pin (15) fully clockwise, then fully counterclockwise and confirm that the FCU maximum stop is contacted by the adjusting screw in both positions. Adjust control rod (13) until the stop is contacted by the adjusting screw. Remove control rod.

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CAUTION: MAKE SURE THAT THE THREADED SECTION OF THE ROD END CONNECTORS (62) EXTEND MORE THAN SAFETY HOLE IN THE FCU INTERCONNECT ROD (13).

9 Tighten locknuts on control rod, ensuring that rod end connectors are aligned with each other, torque to 32 to 36 lb.in., and lockwire. Apply light film of grease (PWC04-001) to ball end fittings of control rod. Install control rod and secure with bolts, washers and castellated nuts. Tighten nuts, torque to 10 to 15 lb.in., and lock with cotterpins.

(h) For PT6A-27 and PT6A-28 Engines, adjust length of starting flow control rod (8):

1 Set FCU arm (10) to approximately 16 degrees below horizontal (Ref. Subpara. (g)2, preceding). Set starting control lever (7) to the RUN position.

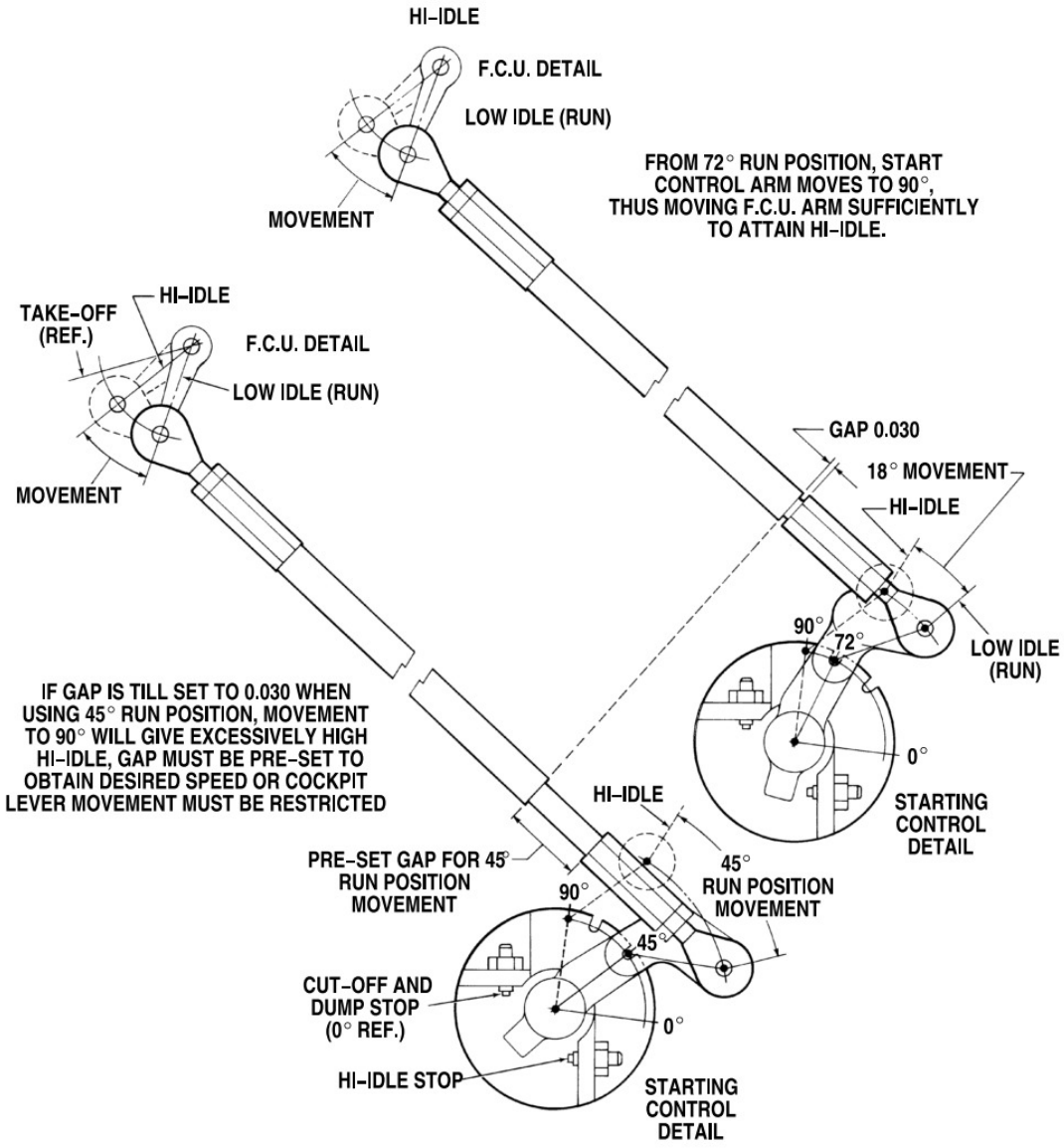
NOTE: Depending on installation, RUN position may be referred to as IDLE, LO-IDLE or GROUND IDLE.

2 Adjust length of control rod to provide a minimum gap of 0.030 inch at lower end of rod when rod is connected between FCU arm and starting flow control lever (Ref. Detail E).

NOTE: The RUN position (Ref. Fig. 202) on the starting control may be at 45° or 72°, depending on the installation, and is defined by detents or rigging slots in the plate behind the lever. Movement from 45° to 90° has no effect on starting control operation and is provided only to actuate the FCU control arm via the telescopic rod (8) and introduce the hi-idle function. Most installations incorporating this function use the 72° RUN position. With the rod adjusted as above, movement of the start control lever from 72° to 90° will provide approximately the correct hi-idle speed, final adjustment being made by the starting control maximum stop. However for installations using the 45° run position, movement of the start control lever from 45° to 90° will provide a hi-idle speed considerably higher than required. This must be compensated for by increasing the gap at the lower end of the rod as required to obtain the desired rpm or, alternatively, limiting movement of the cockpit starting control lever when selecting FLIGHT-IDLE.

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Figure 202 Adjustment of Starting Control Telescopic Linkage



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3 Tighten rod end connector locknuts, torque to 32 to 36 lb.in., and lockwire.

4 Apply light film of grease (PWC04-001) to ball end fittings of control rod, install control rod and secure with bolts, washers and castellated nuts. Tighten nuts, torque to 12 to 18 lb.in., and lock with cotterpins.

(4) Install the wire rope casing:

(a) Install the rear swivel joint (19) on the rear fireseal and temporarily attach with two locknuts.

(b) Install the seals, insulation and seal retaining plates (25) and the bracket (26) on the center fireseal, with the bracket on the front face of the fireseal. Loosely attach with the bolts and the self-locking nuts. The bolt heads to be on air inlet side of the fireseal.

(c) Install the wire rope casing (20) through the center fireseal and connect the coupling nuts (53) and (54) of the casing to the front and rear swivel joints. Adjust position of the rear swivel joint to compensate for variations in the casing length. Tighten the coupling nuts of the casing, torque to 40 to 60 lb.in., and lockwire.

(d) Attach the rear swivel joint to the rear fireseal and support the bracket. Tighten the locknut, torque to 80 to 100 lb.in., and lockwire.

(e) Tighten the self-locking nuts, at the center fireseal and tighten until firm contact is obtained, plus 180 degrees.

(f) Install the loop clamp (58) on the casing and attach to the bracket (59) with a bolt and nut at flange C. Torque the nut 36 to 40 lb.in.

(g) Install the loop clamp (23) on front section of the casing and attach to the bracket (26) with a bolt and self-locking nut. Tighten the nut and torque 36 to 40 lb.in.

(h) Install the loop clamp on rear section of the casing and attach to the similar clamp on T5 trim harness with a bolt and the self-locking nut. Tighten the nut and torque 36 to 40 lb.in.

(i) Insert the wire rope (21) into the rear swivel joint (19) and push forward through the casing. Check the clamping bolt (37) at front terminal is loose and screw the wire rope clockwise to engage internal thread of the clamping bolt. Continue turning the wire rope until the rope bottoms. Check by inserting the wire in inspection hole of the terminal.

(j) Attach the wire rope and the clamping bolt with a self-locking nut. Tighten the nut and torque 10 to 15 lb.in.

(k) Install the rear wire rope terminal (2), complete with the rod end clevis (1) on the wire rope (21) and the rear swivel joint. Rotate the terminal clockwise to engage the wire rope and internal thread of the clamping bolt (3). Check that the wire rope is fully engaged by inserting the wire in inspection hole of the terminal. Tighten the clamping bolt self-locking nut and torque 10 to 15 lb.in.

NOTE: Inspection hole is located adjacent to the rod end clevis on some engines, or on hexagon section of the terminal (Post-SB1170).

(l) Release the locknut that attaches the rod end clevis (1) to the terminal (2). Rotate the

clevis to align with center hole of the propeller control cam (16). Lengthen the wire rope assembly by rotating the clevis one-half turn out to ensure positive movement of the control linkage. Tighten the locknut, torque 75 to 85 lb.in., and safety wire.

(m) Locate the rod end clevis on the control cam and attach to center hole with the straight headed pin (6), the washer and the cotterpin.

CAUTION: DO NOT SELECT CONTROLS INTO REVERSE CONDITION WITHOUT DISCONNECTING FRONT ROD END CLEVIS FROM THE PROPELLER REVERSING LEVER. FREEDOM OF MOVEMENT IN REVERSE CAN ONLY BE DONE WITH ENGINE RUNNING.

(n) On completion of installation and rigging, operate controls and check linkage for freedom of movement and ensure all fittings are secure and locked.

6. Cleaning/Painting

Clean all parts with solvent (PWC11-027) or (PWC11-031) and dry with filtered compressed air and/or clean, lint-free cloths.

7. Inspection/Check

A. General

(1) Visually inspect the actuating levers, control cam, and control lever mounting bracket for cracks. Cracks are not permitted.

(2) Visually inspect the reversing lever guide pin bracket assembly on both faces of the fillet weld area around the guide pin for cracks. If cracks are found and/or fillet weld is on one face only, replace the bracket assembly (Ref. SB1416).

(3) Examine the parts for minor surface damage such as scores, nicks, scratches, and gouges. Minor surface defects may be cleaned up by local blending with a fine stone or crocus cloth (PWC05-061) making sure that all sharp edges and high spots are removed.

(4) Inspect the pin and bolt holes in each part and the slot in the control cam, for burrs and/or chipped edges. Clean up minor damage with a suitable Swiss file or fine stone.

(5) Examine the grooved pin for burrs, nicks, scores and similar defects on serrations which might interfere with the fit between mating parts. Clean up minor defects with a fine stone and check the fit of mating parts.

(6) Inspect the flexible shaft (wire rope) casing for cuts, dents, cracks, kinks, and similar defects which might prevent free movement of the internal wire rope. Minor damage is acceptable, providing the wire rope will operate freely without binding.

(7) Check the wire rope for breaks in the wires, twisting, kinking, and for separation of the reverse wound coils. Reject the wire rope if damaged. Repair is not permitted.

B. Spring Inspection Requirements

(1) Check the extension spring (18, Fig. 201) for distortion and uniformity of the coil spacing. If the coils are unevenly separated in a free state, reject the spring. Minor, even separation of the coils is acceptable providing the requirements of Step (2) following, are satisfied.

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(2) Measure the extension loading on the spring. The load required to extend the spring to a length of 3.70 inches must not be less than 2.25 pounds.

(3) Examine the spring to make sure that the surfaces are free of imperfections such as pits, nicks, and other defects which may be detrimental to fatigue resistance of the spring.

8. Adjustment/Test

A. Procedure

(1) The adjustments detailed in Para. 5. B.(2)(g) are designed to provide the correct power relationship at all ambient temperatures. It is assumed that all rigging is accomplished with the propeller installed and blades in full feather position.

(2) As aircraft installations differ greatly, reference must be made to the applicable Aircraft Maintenance Manual for final adjustment and test.

9. Fits and Clearances

A. General

The following provide the fits and clearances, torques, spring pressures, special assembly, and backlash checks where applicable.

B. Dimensional Checks

The DIMENSIONS FOR REF. column indicates minimum and maximum manufacturing dimensions of two mating parts. These dimensions are provided for information only.

The LIMITS column indicates the desired minimum and maximum fits and clearances between new parts, and also the allowable limit to which these parts may wear before replacement is necessary.

The letter T indicates a tight fit; L or no letter, a loose fit.

BY SELECTION means parts must be matched by selection to provide a required fit.

FIT TO means a fitting operation may be required at assembly to obtain the required fit.

An asterisk (*) indicates part(s) should be replaced if any looseness is evident.

Unless otherwise stated, all fits are diametrical. Spline fits are calculated from chordal dimensions.

C. Torque Limits

Unless otherwise stated, apply thread lubricant to all parts that are to be torque-loaded.

Use engine lubricating oil or equivalent, unless otherwise specified.

Where torque limits for castle nuts are provided in the MINIMUM column only, these nuts should be tightened to the designated torque and then further tightened, if necessary, to properly align the locking slots and holes.

Where MINIMUM and MAXIMUM values are given, the alignment of the locking slot must be obtained without loosening the nut and without exceeding the maximum limit. If this is not

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possible, back-off the nut half a turn, then retighten. If alignment cannot be accomplished, select another nut.

Torque requirements for interference fit applications, such as studs, may be obtained with or without lubrication, unless otherwise specified.

Table 201 Engine - Fits and Clearances

REF. No.	Name	Dim. for Ref. inches		Limits inches		
		Min.	Max.	Min.	Max.	Replace
2509 (**Post-SB1797**)	Internally Threaded Reset Arm Adjuster	0.3440	0.3475	0.0030	0.0100	
	Reset Arm Adjuster	0.3375	0.3410			

Table 202 Torque and Stretch

REF. No.	Name	Limits	
		Min.	Max.
NOT APPLICABLE			

Table 203 Spring Pressures

REF. No.	Name	Limits	
		Min.	Max.
NOT APPLICABLE			

Table 204 Special Assembly Procedures

REF. No.	Name	Limits	
		Min.	Max.
1344 (**Post-SB1797**)	Internally Threaded Reset Arm Adjuster Reset Arm Adjuster		
1345 (**Post-SB1797**)	Rod End Connector		

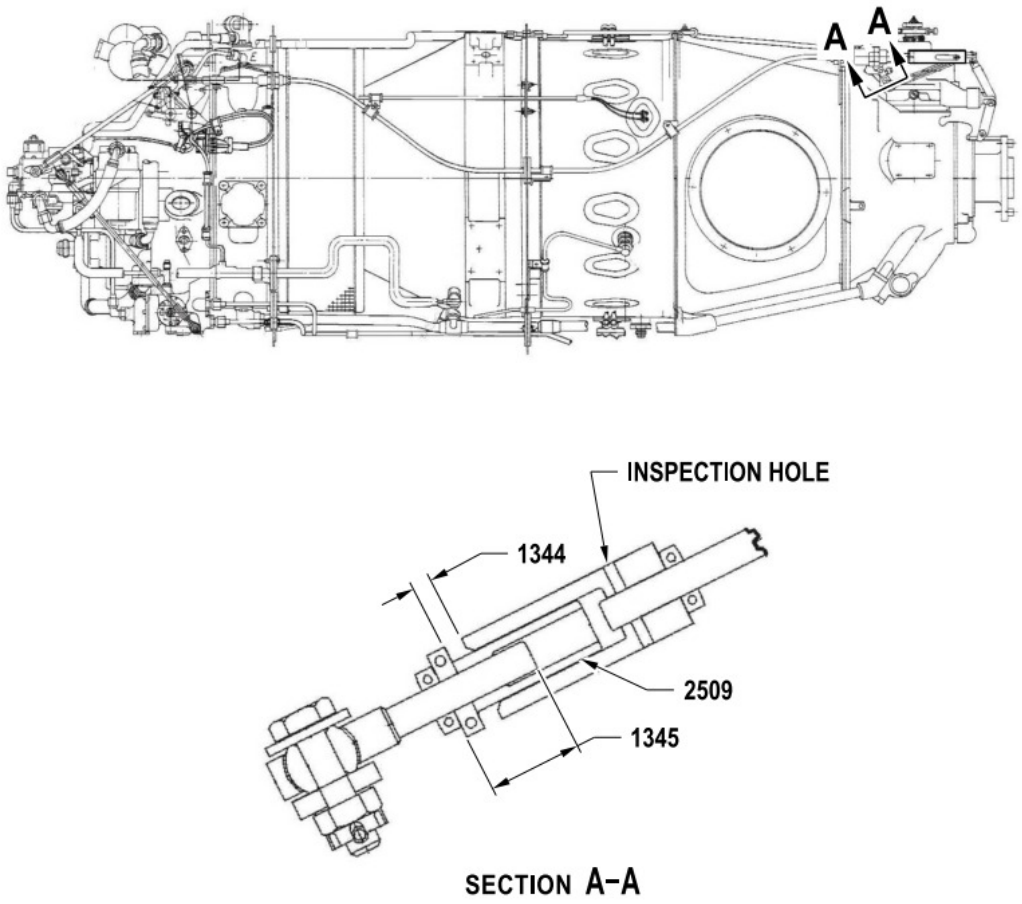
Table 205 Backlash

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REF. No.	Name	Limits	
		Min.	Max.
NOT APPLICABLE			

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Figure 203 Fits and Clearances



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